

Vito Secona

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SKILLS

Skills: Web Development, JavaScript/TypeScript, Rust, Go, C, C++, MLIR

Languages: Bahasa Indonesia (native), English (fluent)

Interests: Software Engineering, Compilers, High Performance Computing

EXPERIENCES

Undergraduate Student Researcher | *Fakultas Ilmu Komputer, Universitas Indonesia* Jan. 2026 — Present

- Main topics: Sparse Tensor Compilers and HPC.
- Investigate MLIR internals with the goal of contributing upstream.
- Architectural analysis of the SparseTensor dialect on MLIR.
- Target outcome: a publishable research paper by Summer of 2026.

GSoC 2025 contributor, The Rust Foundation | *Google Summer of Code* Jul. 2025 — Sep. 2025

- Advancing Rust's build ecosystem by contributing to the Cargo Plumbing Commands project as part of Google Summer of Code 2025.
- Improving Cargo by focusing on decoupling build steps and exposing critical low-level processes.
- Maintaining the `cargo-plumbing` package at `nixos/nixpkgs`.

Teaching Assistant | *Fakultas Ilmu Komputer, Universitas Indonesia* Jun. 2024 — Present

- Assisting the Advanced Programming course.
- Assisting Introduction to Computer Organization course.
- Assisted Programming Foundations 1 course, helping students get started with Computer Science using the Python programming language.
- Assisted Programming Foundations 2 course, helping students understand OOP concepts and problem solving skills.
- Assisted Introduction to Digital Systems course, helping students understand logic circuits and digital design.

Lead Web Developer | *Open House Fasilkom UI 2024* Jun. 2024 — Nov. 2024

- Led the development of a web platform that handled 2000+ users with 1000+ online tickets registrations.
- Developed participant and ambassador registration systems with an efficient workflow.
- Optimized service to handle 100+ concurrent users, ensuring user experience.

Backend Developer | *Green Welfare Indonesia* Jun. 2024 — Jan. 2025

- Drove a 10% performance gain and enhanced overall platform stability by engineering an optimized backend infrastructure using Go, Gin, and Gorm.

OPEN SOURCE PROJECTS

Cargo Plumbing Commands | <https://github.com/crate-ci/cargo-plumbing>

- Exposing core Cargo functionalities as distinct, scriptable steps.
- Easier maintainability and contributor approachability by isolating Cargo's architecture.
- Enabling seamless integration with external build systems (e.g., Bazel, Nix) by refactoring Cargo's architecture to decouple core processes.
- Currently vetting a long-standing system architecture design, leading to a more efficient usage of the Index.

Belalang Programming Language | <https://github.com/belalang-project/belalang>

- A custom programming language designed that features a compiler and virtual machine written in Rust.
- Developed a CLI tool to compile and execute bytecode, with a structured language pipeline including a lexer, parser, and compiler.
- Implemented a stack-based virtual machine for efficient execution.
- Improving memory footprint and performance by implementing the Immix Garbage Collection algorithm.

TryGlasses | <https://github.com/secona/tryglasses-web>

- Simulates wearing eyewear with advanced 3D rendering techniques powered by WebGL.
- Leverages a machine learning-based model for head reconstruction from a single portrait image.
- Incorporates an efficient pipeline to augment the input image with selected eyewear.
- Delivers a realistic and immersive try-on experience using state-of-the-art technology.

EDUCATION

Undergraduate Computer Science | *Universitas Indonesia*

Aug. 2023 — Jun. 2027

Current GPA: 3.92

- Computer Graphics: Rasterization Algorithm, Ray Tracing Algorithm, Deep Learning in Computer Graphics, Animations
- Operating Systems: GNU/Linux CLI, Scripting, Memory Management, Processes and Threads, Synchronizations, File Systems, CPU Scheduling
- Embedded Systems: Real-Time Operating Systems, Arduino Programming, AVR Programming
- Computer Networks: OSI Model, Network Topology Design, Socket Programming
- Algorithms Design and Analysis: Algorithms Correctness and Complexity, Dynamic Programming
- Numerical Analysis: Numerical Stability in Computers, Optimization Techniques in Machine Learning

ACHIEVEMENTS

Beasiswa Bakti BCA - BCA

Jan. 2025 — Dec. 2025

- Awarded based on academic achievements.

CERTIFICATIONS

English Proficiency Test (EPT) - *Universitas Indonesia, LBI*

Sep. 2023

- Achieved a score of 620, equivalent to standardized TOEFL® scores.